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Professional Summary

Computer Science & Engineering graduate (**B.Tech, 9.1/10.0 CGPA**) with strong conceptual and practical grounding in low-level system programming, algorithm design, discrete mathematics, and Linux system administration.

Secured **All India Rank 2098** (GATE Score: 644/1000) out of **211,020 national candidates** in **GATE 2026 (CS & IT)** and **AIR 392** in **IIITH PGEE**.

Experienced in writing memory-efficient C applications, developing custom browser extensions (Manifest V3), automating workflows on Fedora Linux Workstation via Python & Shell scripts, and building cloud automation pipelines.

Demonstrates strong technical ownership, meticulous attention to detail, ethical work standards, and a continuous drive for engineering excellence.

Education

B.Tech Rajasthan Technical University, Computer Science & Engineering

Jaipur, Rajasthan

Aug 2022 – May 2026

- **CGPA:** 9.1 / 10.0
- **Relevant Coursework:** Data Structures & Algorithms, Discrete Mathematics, Operating Systems, Database Management Systems (DBMS), Computer Networks, System Programming, Microprocessors, Linear Algebra, Abstract Algebra (Group Theory).
- **Academic Standing:** Top-tier academic performance with a rigorous mathematical and theoretical foundation across core CS domains.

Competitive Examinations

- **GATE 2026 (Computer Science & Information Technology): All India Rank: 2098** out of **211,020 registered candidates** (Top ~0.99% nationwide) | **GATE Score: 644 / 1000** | **Registration No: CS26S33025126**. Demonstrated comprehensive problem-solving mastery in Data Structures, Algorithms, Discrete Mathematics, OS, DBMS, Theory of Computation, Digital Logic, and Computer Architecture. (*Official Scorecard Attached*).
- **IIITH PGEE (IIIT Hyderabad Postgraduate Entrance Exam): All India Rank: 392** nationwide out of ~10,000–15,000 post-graduate engineering candidates.

Certifications

- **NPTEL Programming in Java:** Awarded **Elite Certificate** with **80% Score** | Ranked among top national performers out of 14,693 candidates (Jan–Apr 2024).
- **NPTEL Problem Solving Through Programming in C:** Awarded **Elite Certificate** with **73% Score** out of 3,749 certified candidates (Jan–Apr 2023).
- **NPTEL Programming in Modern C++:** Awarded **61% Final Score** | Completed 12-week national course covering C++ specifications, generic programming, and STL architecture out of 1,028 certified candidates (Jul–Oct 2023).
- **Salesforce Certified AI Associate:** Credential ID: 6127333 (Issued May 2025) — Covers AI fundamentals, ethical AI practices, and CRM AI concepts.
- **Salesforce Certified Associate:** Credential ID: 6030076 (Issued April 2025) — Validates platform architecture, user administration, and data management.
- **Building with Google Gemini (Advanced & Ultra):** LinkedIn Learning | Certificate ID: 1df85a0a... (Issued Aug 2025) — Generative AI prompt engineering, Gemini API SDKs, and software integration.
- **Raspberry Pi Essential Training (2022):** LinkedIn Learning | Certificate ID: 20c6cecc... (Issued Sep 2025) — Single-board hardware configuration, Linux GPIO interaction, and IoT setup.
- **C Programming For Beginners – Master the C Language:** Udemy (Jan 2024, 25.5 hours intensive coursework).
- **HACK-AIETM-2024 Hackathon Certificate of Participation:** Issued by Arya Group of Colleges & AIETM, Jaipur (May 2024).

Detailed Technical Skills

Discrete & Foundational Math: Deep conceptual and problem-solving mastery in **Discrete Mathematics** (Mathematical Logic, Propositional & Predicate Calculus, Set Theory, Relations & Functions, Combinatorics, Graph Theory, and Recurrence Relations), **Linear Algebra**, and **Group Theory** (Abstract Algebra). Applied mathematical proof techniques and discrete structures directly to algorithmic analysis, gate logic verification, and GATE CS problem solving.

C & Modern C++: **Theoretical & Core Concepts:** Object-Oriented Programming (OOP) paradigms (polymorphic class hierarchies, inheritance, encapsulation), custom data structures, manual memory management (`malloc/free`, pointers, stack allocation), templates, and operator overloading. **Practical Usage:** Developed low-level parsing engines in C (Infix-to-Postfix converter and stack-based arithmetic evaluator) and backtracking algorithms (Sudoku Solver engine).

Python & Automation Scripting: **Theoretical & Practical Expertise:** Proficient in standard library modules, file handling, regular expressions, JSON parsing, PyDrive/Google Drive API integration, and Telegram Bot API. **Practical Usage:** Built cloud backup automation pipelines (`telegram-to-drive`), web scraping tools (`BeautifulSoup`), batch downloaders, logic evaluation tools, and task management CLI utilities.

Browser Extension Engineering: **Practical Usage (Manifest V3 & JavaScript):** Developed custom Chromium extensions (`ytmaster` / `FocusTube`, `webtools`) to enhance web navigation, automate hotkey search focusing, redirect image search providers, and eliminate video distractions.

Linux Systems Administration: **Practical Usage (Fedora Workstation):** Operating **Fedora Linux Workstation** as primary personal OS. Engineered custom Bash automation scripts and cron jobs for system maintenance, backup execution, desktop environment workflows, and terminal productivity (`zsh`, `Vim/Emacs`, `grep`, `find`, `sed`, `awk`).

Database Management & SQL: **Academic Knowledge & Fundamentals:** Solid theoretical understanding of Relational Database Management Systems (RDBMS), Entity-Relationship (ER) modeling, 1NF to 3NF/BCNF normalization, Relational Algebra, ACID properties, and basic SQL query execution.

Developer Tools & Workflows: **Practical Usage:** Daily usage of `Git` and `GitHub` for version control, `Typst` for typesetting technical documentation, markdown vault notes, and modal text editors.

Projects

GATE Exam Mock Test Interface & Portal Engine

Web Application | JavaScript, HTML5, Local Engine

- Built a custom offline-first mock test interface mirroring the official GATE examination environment for personal exam practice and simulation.
- Implemented interactive virtual keypad controls, question palette status navigation (Attempted, Marked for Review, Unattempted), countdown timer management, and real-time score analytics.
- Enabled zero-latency offline test execution without relying on external server connectivity.

Automated Test Series Data Pipeline & Archival Engine

Data Engineering & Archival | Python, JSON Data Pipeline

- Developed an automated data extraction and JSON indexing pipeline to structure and archive high-volume test series content for offline revision.
- Built automated scripts (`download_all_tests.py`, `redownload_all_tests.py`) for batch processing, image asset caching, and structured schema formatting.

PyQHub – University Semester PYQ Platform

Web Application | TypeScript, Web Architecture

- Developed a web repository platform to accumulate, organize, and search previous year semester exam papers across technical universities (RTU, AKTU, and others).
- Implemented structured course/subject paper categorizations and fast search filtering to help university engineering students prepare for semester examinations.

WebTools Browser Extension

Chrome Extension | Manifest V3, JavaScript, Chrome Extension API

- Engineered a Manifest V3 Chromium browser extension to optimize daily web browsing and search workflows.
- Implemented global hotkey search bar focusing (instant `/` shortcut), automatic Brave Images to Google Images redirection, and popup configuration management via Chrome Storage API.

FocusTube (YT Master) Distraction-Free Extension

Chrome Extension | JavaScript, DOM Manipulation, YouTube API

- Built a specialized YouTube productivity extension designed to eliminate recommendation algorithms, hide feed clutter, and provide customizable UI element toggles.
- Utilized Content Scripts injected at `document_start` and background Service Workers for zero-flicker UI modification.

Telegram to Google Drive Automated Cloud Pipeline

Cloud Automation & System Scripting | Python, Telegram API, PyDrive

- Architected an automated cloud ingestion pipeline in **Python** that intercepts media files and document attachments from Telegram channels and streams them directly into **Google Drive** storage.
- Integrated Google Drive OAuth2 authentication and PyDrive API to manage file upload chunks, remote directory creation, and duplicate file prevention.

Infix to Postfix Converter & Expression Calculator Engine

Systems & Data Structures | C Language, Memory Management

- Designed and implemented a low-level mathematical expression parsing engine in **C** that converts arithmetic expressions from Infix notation to Postfix (Reverse Polish Notation) using custom Stack structures.
- Built expression evaluation logic handling full operator precedence (parentheses, exponentiation, multiplication, division, addition, subtraction).

Boolean Truth Table & Logic Gate Generator

Computer Logic & Discrete Math | Python, Logic Gates

- Developed an automated Boolean logic truth table evaluator capable of parsing arbitrary logical expressions (AND, OR, NOT, XOR, Implication).
- Dynamically evaluated truth values across all 2^n variable combinations and formatted structured tabular outputs for digital logic verification.

Sudoku Solver Engine

Algorithms & Backtracking | C++, Constraint Satisfaction

- Built an efficient 9x9 Sudoku puzzle solver using recursive **Backtracking** algorithms and constraint satisfaction logic in **C++**.

TODO CLI & Terminal Task Manager

CLI Utility | Python, Linux File I/O, JSON

- Developed a lightweight command-line task management tool in **Python** tailored for fast Linux terminal execution without GUI overhead.

Fedora Linux Dotfiles & System Workflows

DevOps & System Configuration | Bash, Shell, Custom Configs

- Maintained a modular repository of personal **Fedora Linux Workstation** dotfiles, shell aliases, Vim/Emacs configurations, desktop environment settings, and custom terminal shortcuts.